Problem Statement : Implement a Linear Regression Model to predict house prices for regions in the USA using the provided dataset.

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn import metrics
import warnings

warnings.filterwarnings('ignore')

HouseDF = pd.read_csv('C:/Users/Admin/Downloads/USA_Housing.csv')

HouseDF.head()
```

```
   Avg. Area Income  Avg. Area House Age  Avg. Area Number of Rooms  \
0      79545.458574             5.682861                   7.009188
1      79248.642455             6.002900                   6.730821
2      61287.067179             5.865890                   8.512727
3      63345.240046             7.188236                   5.586729
4      59982.197226             5.040555                   7.839388

   Avg. Area Number of Bedrooms  Area Population         Price  \
0                          4.09     23086.800503  1.059034e+06
1                          3.09     40173.072174  1.505891e+06
2                          5.13     36882.159400  1.058988e+06
3                          3.26     34310.242831  1.260617e+06
4                          4.23     26354.109472  6.309435e+05

                                            Address
0  208 Michael Ferry Apt. 674\nLaurabury, NE 3701...
1  188 Johnson Views Suite 079\nLake Kathleen, CA...
2  9127 Elizabeth Stravenue\nDanieltown, WI 06482...
3                          USS Barnett\nFPO AP 44820
4                         USNS Raymond\nFPO AE 09386
```

```
HouseDF.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 7 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Avg. Area Income              5000 non-null   float64
 1   Avg. Area House Age           5000 non-null   float64
 2   Avg. Area Number of Rooms     5000 non-null   float64
 3   Avg. Area Number of Bedrooms  5000 non-null   float64
```

```
 4   Area Population               5000 non-null   float64
 5   Price                         5000 non-null   float64
 6   Address                       5000 non-null   object
dtypes: float64(6), object(1)
memory usage: 273.6+ KB
```

```
HouseDF.describe()
```

```
       Avg. Area Income  Avg. Area House Age  Avg. Area Number of
Rooms  \
count       5000.000000          5000.000000
5000.000000
mean       68583.108984             5.977222
6.987792
std        10657.991214             0.991456
1.005833
min        17796.631190             2.644304
3.236194
25%        61480.562388             5.322283
6.299250
50%        68804.286404             5.970429
7.002902
75%        75783.338666             6.650808
7.665871
max       107701.748378             9.519088
10.759588

       Avg. Area Number of Bedrooms  Area Population         Price
count                   5000.000000      5000.000000  5.000000e+03
mean                       3.981330     36163.516039  1.232073e+06
std                        1.234137      9925.650114  3.531176e+05
min                        2.000000       172.610686  1.593866e+04
25%                        3.140000     29403.928702  9.975771e+05
50%                        4.050000     36199.406689  1.232669e+06
75%                        4.490000     42861.290769  1.471210e+06
max                        6.500000     69621.713378  2.469066e+06
```

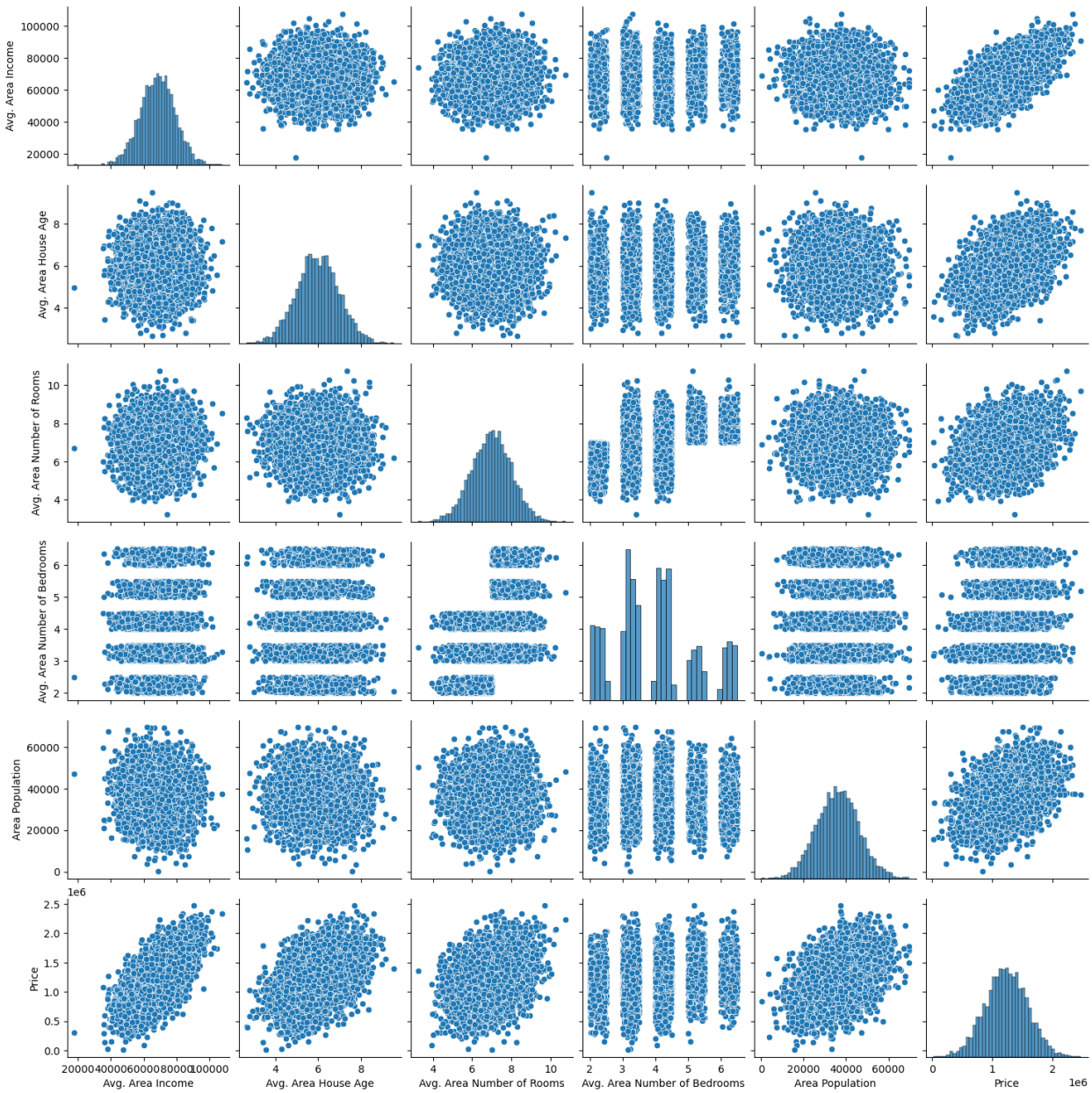
```
HouseDF.columns
```

```
Index(['Avg. Area Income', 'Avg. Area House Age', 'Avg. Area Number of
Rooms',
       'Avg. Area Number of Bedrooms', 'Area Population', 'Price',
'Address'],
      dtype='object')
```

Exploratory Data Analysis for House Price Prediction

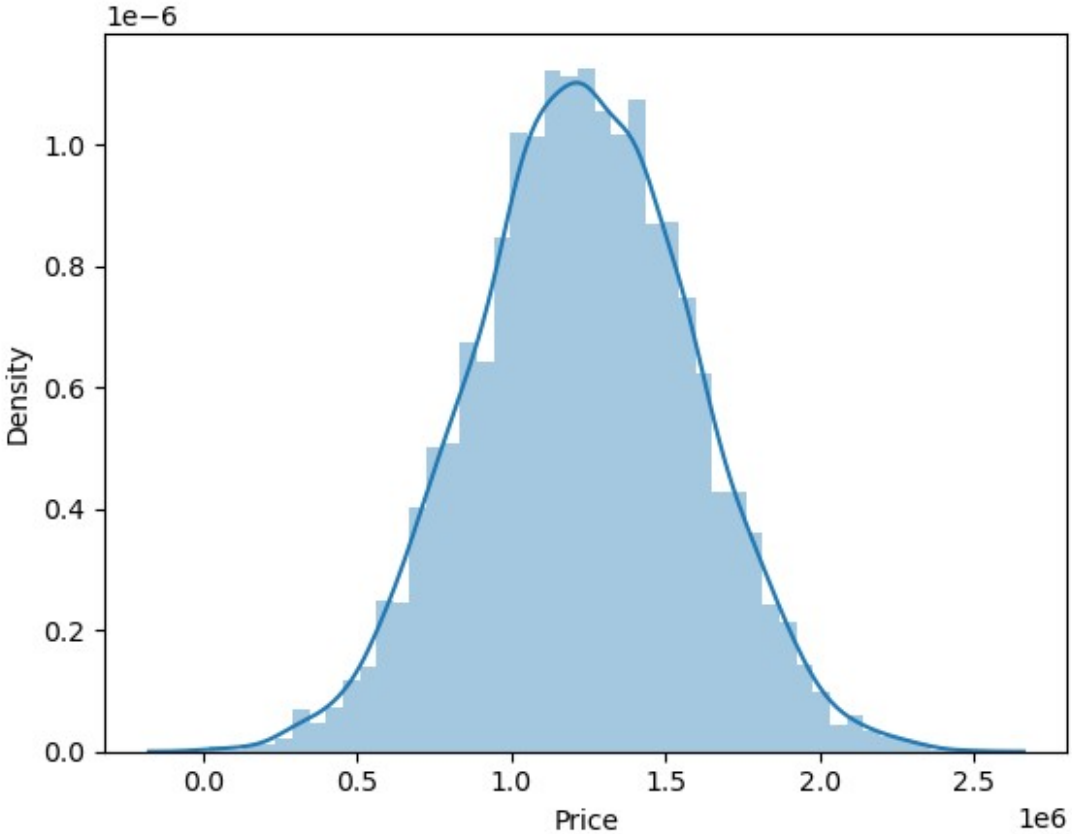
```
sns.pairplot(HouseDF)
```

```
<seaborn.axisgrid.PairGrid at 0x21f940fbe00>
```



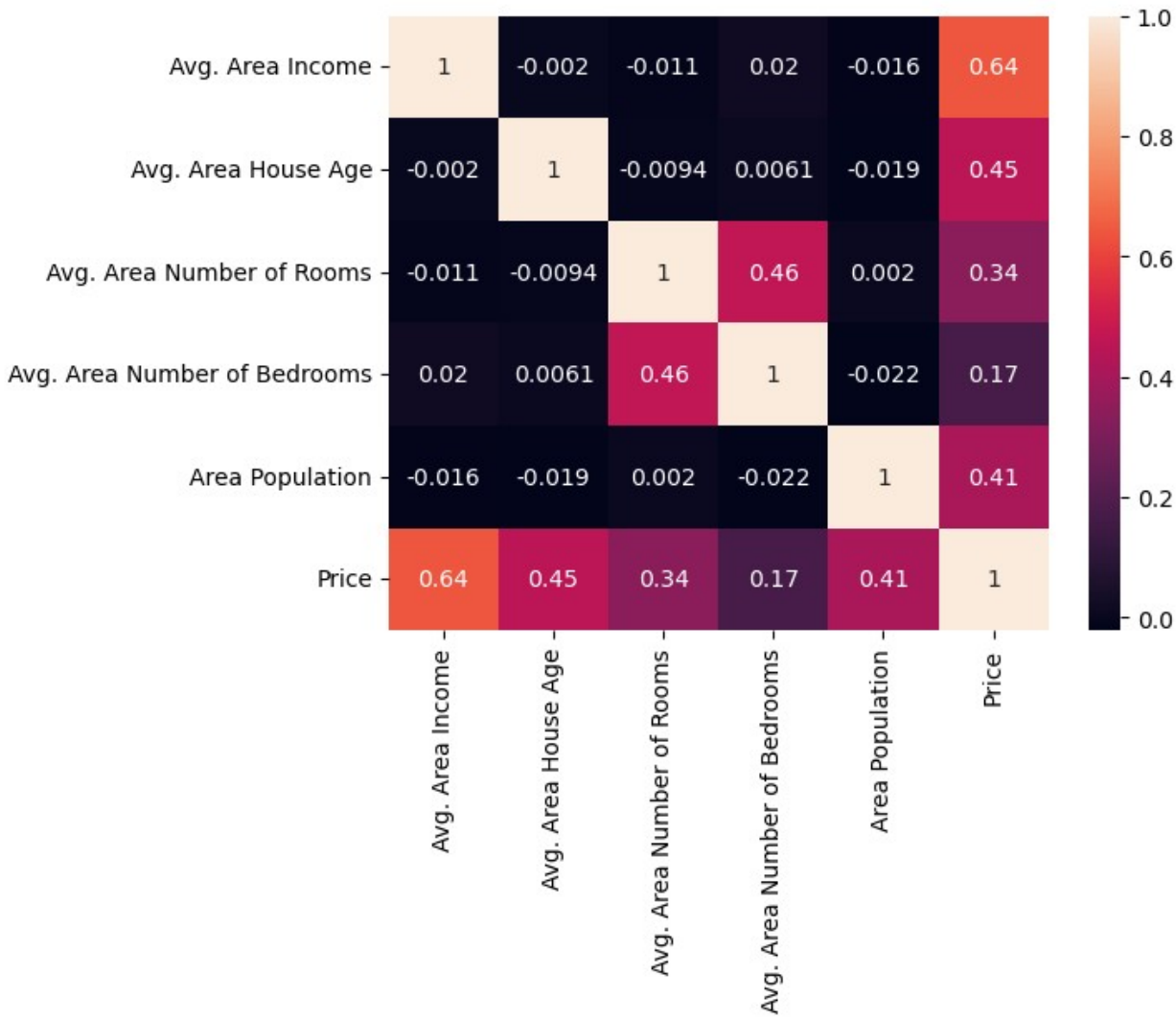

```
sns.distplot(HouseDF['Price'])
```

```
<Axes: xlabel='Price', ylabel='Density'>
```




```
# Select only the columns that have numerical data types
numeric_df = HouseDF.select_dtypes(include=[np.number])

# Plot the heatmap using only the numeric dataframe
sns.heatmap(numeric_df.corr(), annot=True)
```

```
<Axes: >
```



Training a Linear Regression Model

X and y List

```
X = HouseDF[['Avg. Area Income', 'Avg. Area House Age', 'Avg. Area
Number of Rooms',
               'Avg. Area Number of Bedrooms', 'Area Population']]

y = HouseDF['Price']
```

Split Data into Train, Test

```
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.4, random_state=101)
```

Creating and Training the LinearRegression Model

```
lm = LinearRegression()

lm.fit(X_train,y_train)
```

```
LinearRegression()
```

LinearRegression Model Evaluation

```
print(lm.intercept_)
```

```
-2640159.7968525267
```

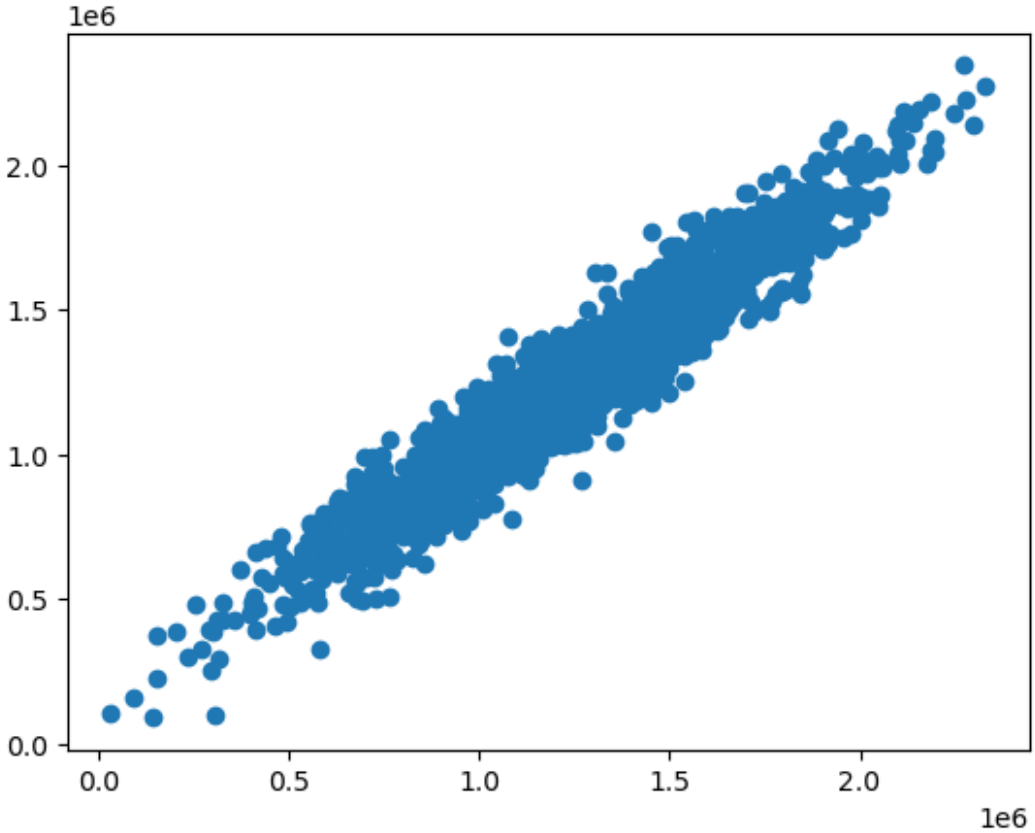
```
coeff_df = pd.DataFrame(lm.coef_,X.columns,columns=['Coefficient'])
coeff_df
```

```
                                  Coefficient
Avg. Area Income                    21.528276
Avg. Area House Age            164883.282027
Avg. Area Number of Rooms      122368.678027
Avg. Area Number of Bedrooms     2233.801864
Area Population                     15.150420
```

Predictions from our Linear Regression Model

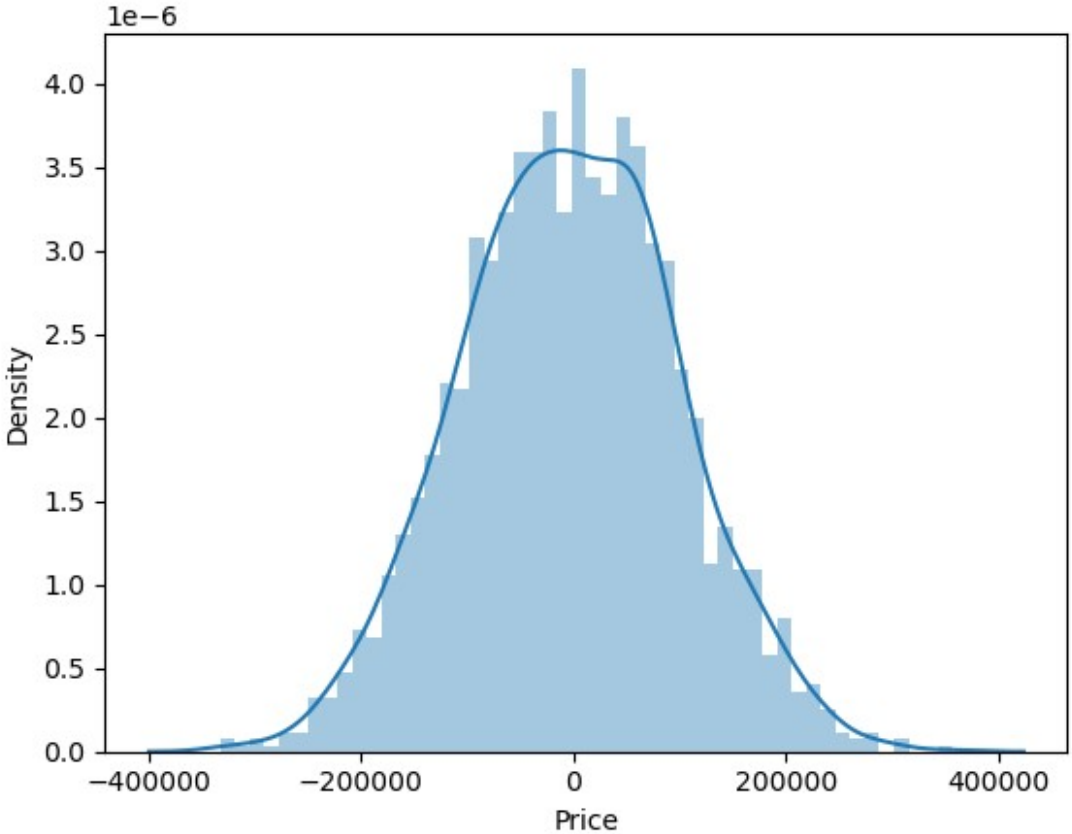
```
predictions = lm.predict(X_test)

plt.scatter(y_test,predictions)
```

```
<matplotlib.collections.PathCollection at 0x21f9c575c40>
```




```
sns.distplot((y_test-predictions),bins=50)
```

```
<Axes: xlabel='Price', ylabel='Density'>
```



Regression Evaluation Metrics

```
print('MAE:', metrics.mean_absolute_error(y_test, predictions))
print('MSE:', metrics.mean_squared_error(y_test, predictions))
print('RMSE:', np.sqrt(metrics.mean_squared_error(y_test,
predictions)))
```

```
MAE: 82288.22251914947
MSE: 10460958907.209059
RMSE: 102278.82922290936
```